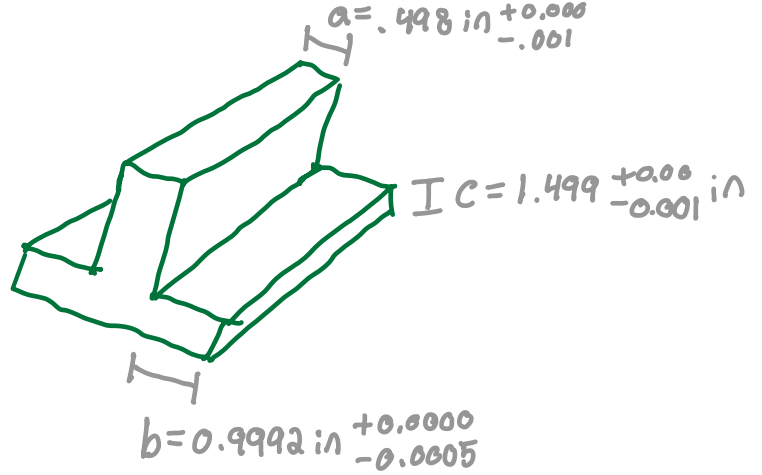
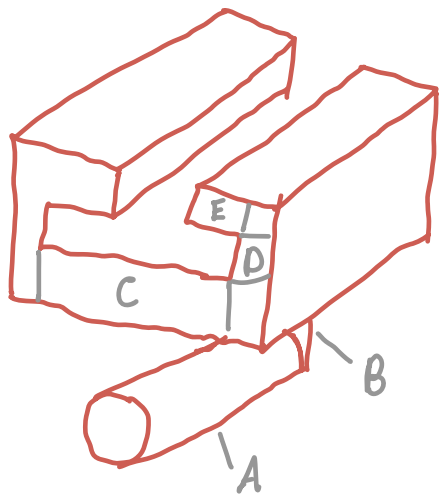




Knowns

$$F = 800 \text{ lbf}$$

$$SS = 4$$

$$\delta_{\max} = 0.005 \text{ in}$$

materials

S_y

σ_{allow}

E

6061-T6

40 ksi

10 ksi

10,000 ksi

A36

36 ksi

9 ksi

29,000 ksi ← I'm going with this one

Ti-6Al-4V

128 ksi

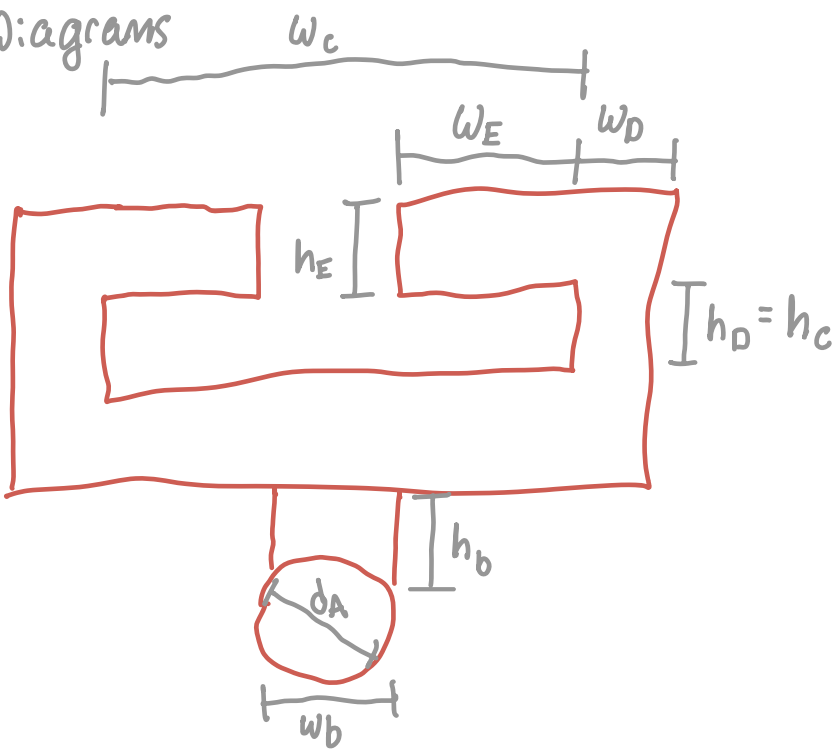
32 ksi

16,500 ksi

Unknowns

Dimensions of members A, B, C, D, E based upon stress from load through members.

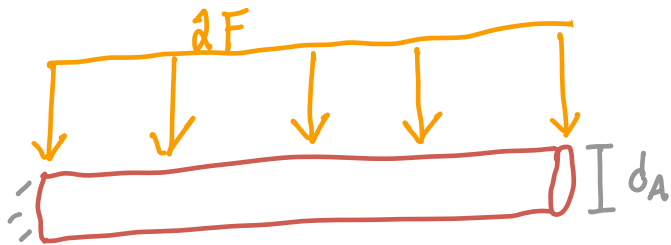
Diagrams



hints

- start with A
- treat A as a cantilever beam
- treat B as axial loaded bar
- C is a simply supported beam w/ a load at the center.

member A



$$Z = \frac{sf \cdot 2F \cdot l}{2 \cdot s_y}$$

$$M_{\max} = \frac{2Fl}{2}$$

$$\sigma_{\max} = \frac{M}{Z}$$

$$d = \sqrt[3]{\frac{32M}{\pi \cdot \sigma_{\text{allow}}}}$$

member A solved

$$Z = \frac{4 \cdot 1600 \cdot 1.5}{2 \cdot 36000} = .1333 \text{ in}^3$$

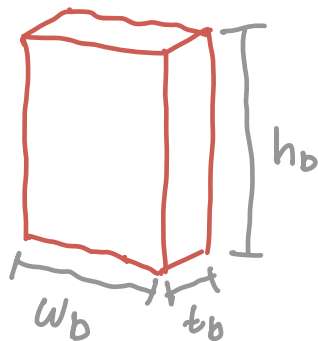
$$M_{\max} = \frac{1600 \cdot 1.5}{2} = 1200 \text{ lb} \cdot \text{in}$$

$$\sigma_{\max} = \frac{1200 \text{ lb} \cdot \text{in}}{.1333 \text{ in}^3} = 9000 \frac{\text{lb}}{\text{in}^2}$$

l = width of strap, 1.5" can handle the load

$$d = \sqrt[3]{\frac{32 \cdot .1333}{\pi}} = 1.1074 \text{ in}$$

member b



$$A = w_b \cdot t_b$$

$$w_b = d_A = 1.1074 \text{ in}$$

$$\frac{2F}{w_b \cdot t_b} = \sigma_{\text{allow}}$$

$$t_b = \frac{2F}{w_b \sigma_{\text{allow}}}$$

$$\delta = \frac{2F h_b}{AE}$$

$$t_b = \frac{2F h_b}{E \cdot w_b \cdot \delta}$$

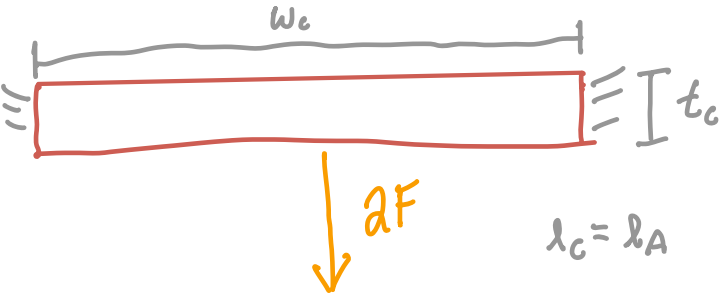
h_b doesn't matter when not accounting for bending, we will leave $\frac{1}{16}$ for the strap, $\frac{1}{8}$ for clearance, and $1.1074/2 \text{ in}$ for the radius so total $\approx .75 \text{ in}$

member b solved

$$t_b = \frac{2(800)}{1.1074 \cdot 9000} = .1605 \text{ in} \leftarrow \text{governing is stress}$$

$$t_b = \frac{1600 \cdot .75}{29 \cdot 10^6 \cdot 1.1074 \cdot .005} = .00747 \text{ in}$$

member c



T-bar flange
T-bar central

$$w_c = 2(.9992) + .498 = 2.5064 \pm .000 \text{ in}$$

F = Force at each end acting downwards

Stress

$$M = \frac{2F \cdot l_c}{4}$$

$$Z = \frac{l_c t_c^2}{6}$$

$$t_c = \sqrt{\frac{3 F w_c}{l_c \cdot \sigma_{allow}}}$$

Stiffness

$$\delta = \frac{2 F w_c^3}{48 E I}$$

$$I = \frac{w t^3}{12}$$

$$t_c = \sqrt[3]{\frac{F w_c^3}{2 E l_c \cdot \delta_{allow}}}$$

member c solved

Stress

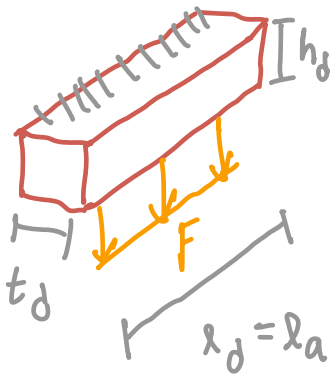
$$t_c = \sqrt{\frac{2400 \cdot 2.5264}{1.5 \cdot 9000}} = .6702 \text{ in}$$

↑
governing

Stiffness

$$t_c = \sqrt[3]{\frac{800(2.5064)^3}{2 \cdot 29 \cdot 10^6 \cdot 1.5 \cdot .005}} = .3095 \text{ in}$$

member d



knowns

$$l_d = 1.5 \text{ in}, h_d = 1.509^{+0.020}_{-0.000}$$

unknown

$$t_d$$

stress

$$\sigma_{allow} = \frac{F}{t_d \cdot l_d}$$

stiffness

$$\delta = \frac{F h_d}{l_d \cdot t_d \cdot E}$$

$$t_d = \frac{F}{\sigma_{allow} \cdot l_d}$$

$$t_d = \frac{F h_d}{l_d \cdot \delta \cdot E}$$

member d solved

stress

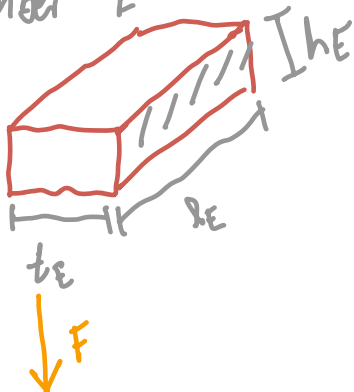
$$t_d = \frac{800}{9600 \cdot 1.5 \text{ in}} = .0593 \text{ in}$$

this is not realistic to machine, I'd just go with .125 in

stiffness

t_d = even smaller

member E



$$l_E = 1.5 \text{ in}$$

$$t_E = .9992 \text{ in}^{+0.000}_{-0.010}$$

stress

similar to member C

$$h_E = \sqrt{\frac{6 F \cdot t_E}{l_E \sigma_{allow}}}$$

member E solved

$$h_E = \sqrt{\frac{6.800 \cdot .9992}{1.5 \cdot 4 \cdot 10^6}} = .596 \text{ in}$$

stiffness is not the ruling dimension.